Case Study 2:

Model Predictive Control of Hemodynamic Variables during Hemodialysis

References

Material Presented in this chapter was taken from:

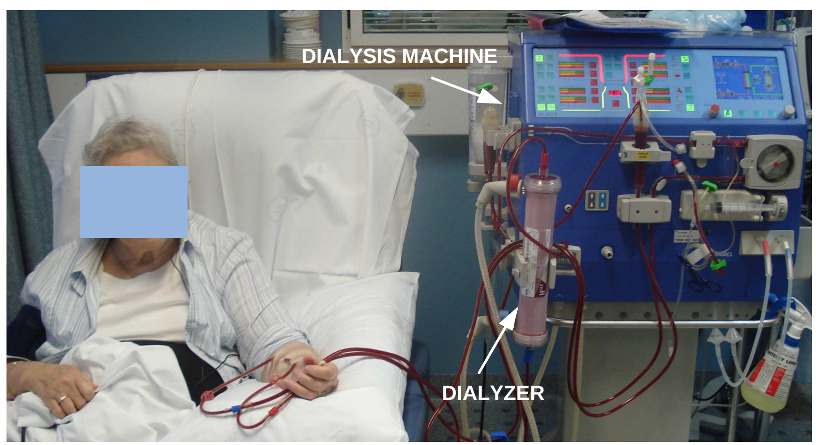
- F. Javed, A. V. Savkin, G. S. H. Chan, et al., "Identification and Control for Automated Regulation of Hemodynamic Variables During Hemodialysis," *IEEE Transactions on Biomedical Engineering*, In Press, 2011.
- F. Javed, A. V. Savkin, G. S. H. Chan, et al., "Model predictive control of relative blood volume and heart rate during hemodialysis," *Med Biol Eng Comput (MBEC)*, **48**: 389-397, 2010.
- F. Javed, G. S. H. Chan, A. V. Savkin, et al., "Assessing the blood volume and heart rate responses during haemodialysis in fluid overloaded patients using support vector regression," *Physiol. Meas*, **30**: 1251-1266, 2009.

Objectives

- To propose a linear time-varying model to describe the response of hemodynamic variables during hemodialysis.
- To develop and test a computer-controlled system to regulate relative blood volume (RBV) and heart rate (HR) while maintaining blood pressure within bounds during hemodialysis.

Introduction

- Kidney failure patients rely on hemodialysis as life sustaining therapy.
- The extra fluid accumulated in the patient's body is removed by the process of diffusion and ultrafiltration.
- The reduction in blood volume lead to relative hypovolemia which can result in sudden drop in blood pressure.
- The stability of patient rely on compensatory mechanisms which include sympathetic vasoconstriction and modest heart rate increase.



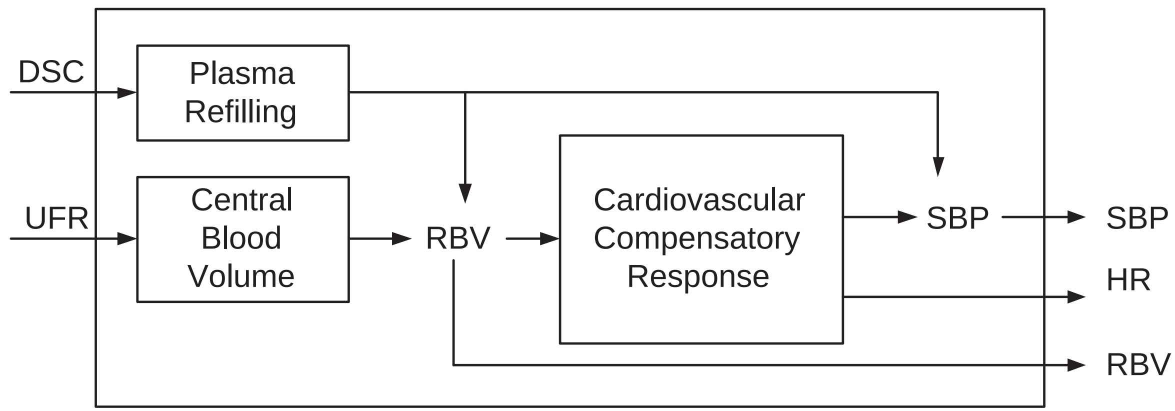
Motivation

- Ultrafiltration Rate (UFR): rate at which fluid is removed in Litres/hour.
- Dialysate sodium concentration (DSC): has a direct impact on blood pressure response.
- During conventional hemodialysis both UFR and DSC are set to constant levels at the start of dialysis session.
- The system operates in open-loop with no feedback of the patient's condition.
- A closed loop system that can adjust the UFR and DSC based on the hemodynamic state of the patient can improve hemodialysis system performance and can ensure the stability of patient.

Modeling

- Part 1: Modeling of hemodynamic variables

Block representation of model



- The UFR and DSC directly influence the blood volume in the intravascular compartment which can be measured as RBV.
- The loss in blood volume activates cardiovascular compensatory response leading to an increase in HR to maintain SBP.
- The SBP response can also be influenced by the change in DSC.

The Proposed Model

$$
\begin{aligned}
\mathbf{x}_{k+1} &= \mathbf{A}_k \mathbf{x}_k + \mathbf{B}_k \mathbf{u}_k + \mathbf{F} + w_k \\
y_k &= \mathbf{C}_k \mathbf{x}_k
\end{aligned} \tag{1}
$$

$$
\mathbf{A}_k = \begin{bmatrix} (1-\xi) & 0 & 0 \\ b_{1,k} & 1 & 0 \\ c_{1,k} & 0 & 1 \end{bmatrix}, \mathbf{B}_k = \begin{bmatrix} \alpha a_{1,k} & -\beta a_{1,k} \\ 0 & 0 \\ 0 & c_2 \end{bmatrix},
$$

$$
\mathbf{F} = \begin{bmatrix} a_2 \\ 0 \\ c_3 x_{3,0} - 0.5 c_2 \end{bmatrix}, \mathbf{C}_k = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{2}
$$

The model (Cont'd)

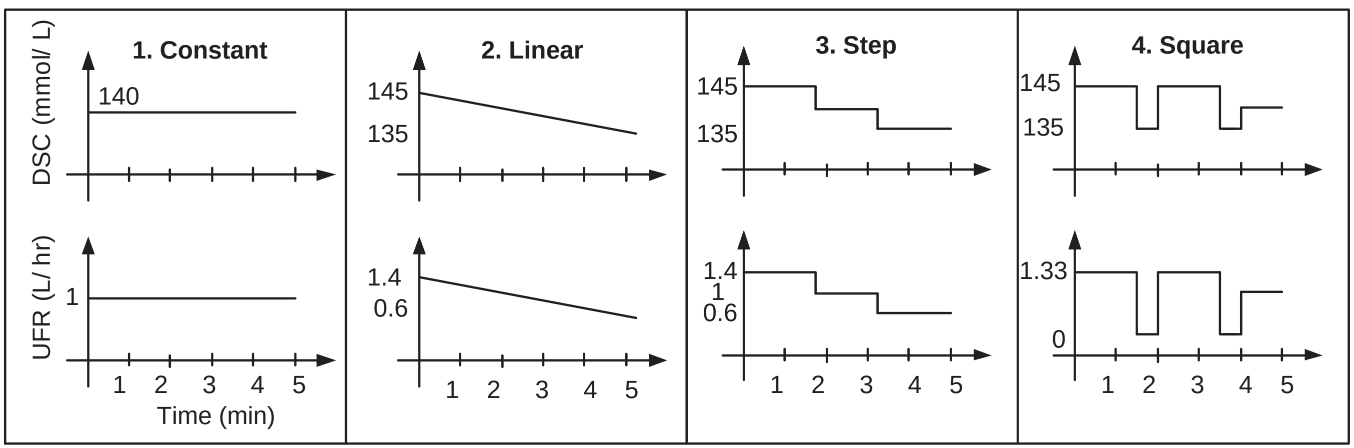
$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{A}_k \mathbf{x}_k + \mathbf{B}_k \mathbf{u}_k + \mathbf{F} + w_k \\ y_k &= \mathbf{C}_k \mathbf{x}_k \end{aligned} \quad (3)$$

where

- States: $x_1(k)$, represent RBV,$x_2(k)$ is HR and $x_3(k)$ is SBP;
- Control Inputs: $u_1(k)$, is the UFR and $u_2(k)$ is DSC.
- Time-varying parameters: $a_{1,k}$, $b_{1,k}$, and $c_{1,k}$.
- Time-invariant parameters: ξ, a_2, c_2 and c_3.
- $x_{3,0}$ is the pre-dialysis SBP.

Parameter estimation

- Subjects: 12 hemodialysis patients were studied.
- Procedure: Each patient underwent start-of-week hemodialysis with each of the four dialysis protocols depicted in Figure below:



Parameter estimation

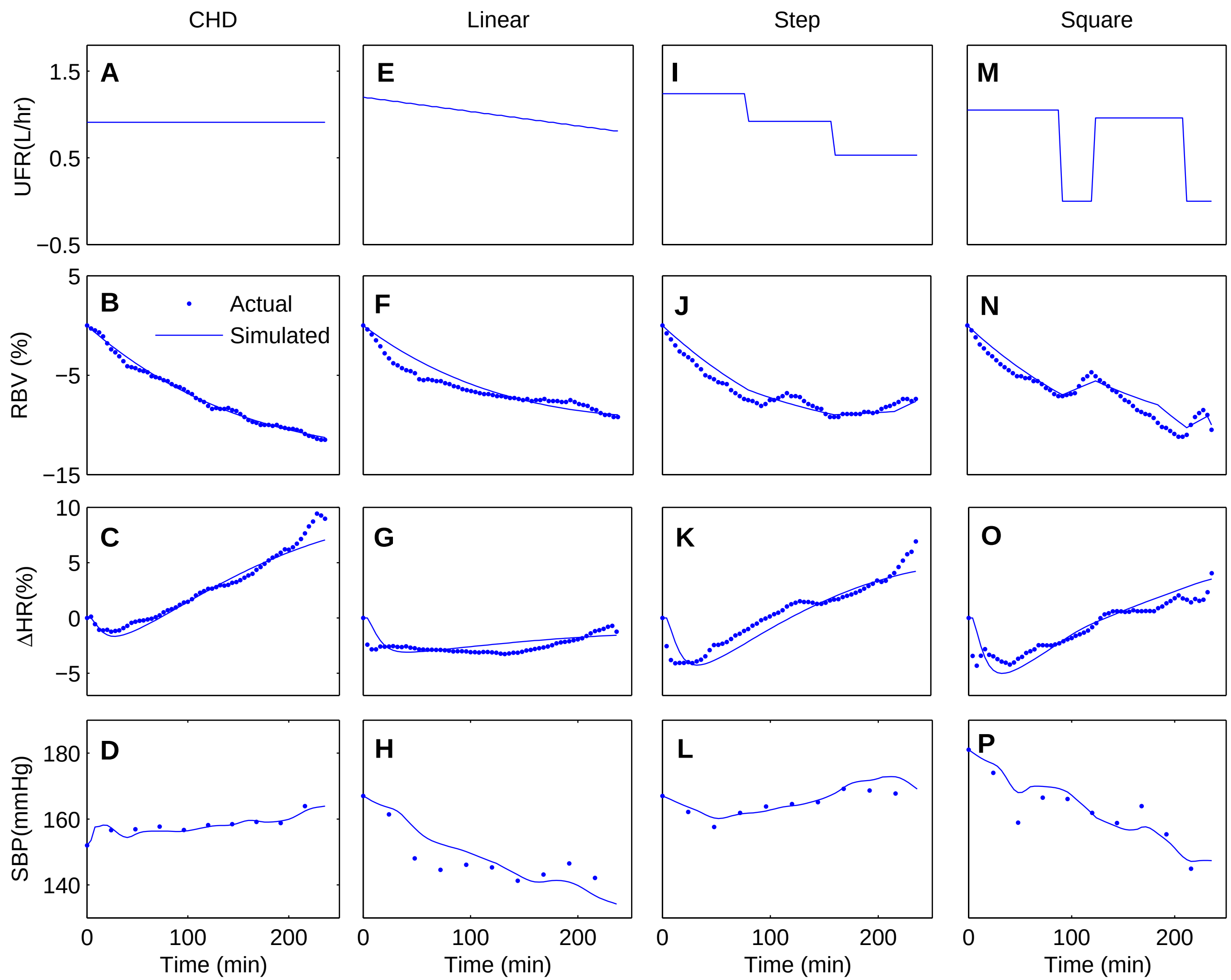
- The model parameters were estimated for an individual patient using data from all four profiled dialysis sessions.
- For the identification of RBV, first the time-invariant parameters ξ and a_2 were estimated based on the refilling response during a square profile.
- Next the time-varying parameters $a_{01,k}$ and $a_{11,k}$ were estimated using the MATLAB curve fitting toolbox based on the mean curves of RBV during all four profiles, assuming that the values of γ, α and β were equal to one.
- The time-invariant parameters γ, α and β were then estimated by tuning the individual patient's response to each profile.

Parameter estimation

- For HR, the time-varying parameters $b_{01,k}$ and $b_{11,k}$ were estimated based on the mean curves of ΔHR over all four profiles by assuming that the values of ζ_1 and ζ_2 were equal to one.
- The time-invariant parameters ζ_1 and ζ_2 were then estimated by tuning the individual patient's response to each profile.
- For SBP, the time-invariant parameters c_2 and c_3 were estimated based on mean curves of SBP over all four profiles and the time-varying parameter $c_{1,k}$ was estimated by tuning an individual patient's response to each profile.

Model response

- Model estimation response for one of the patients.

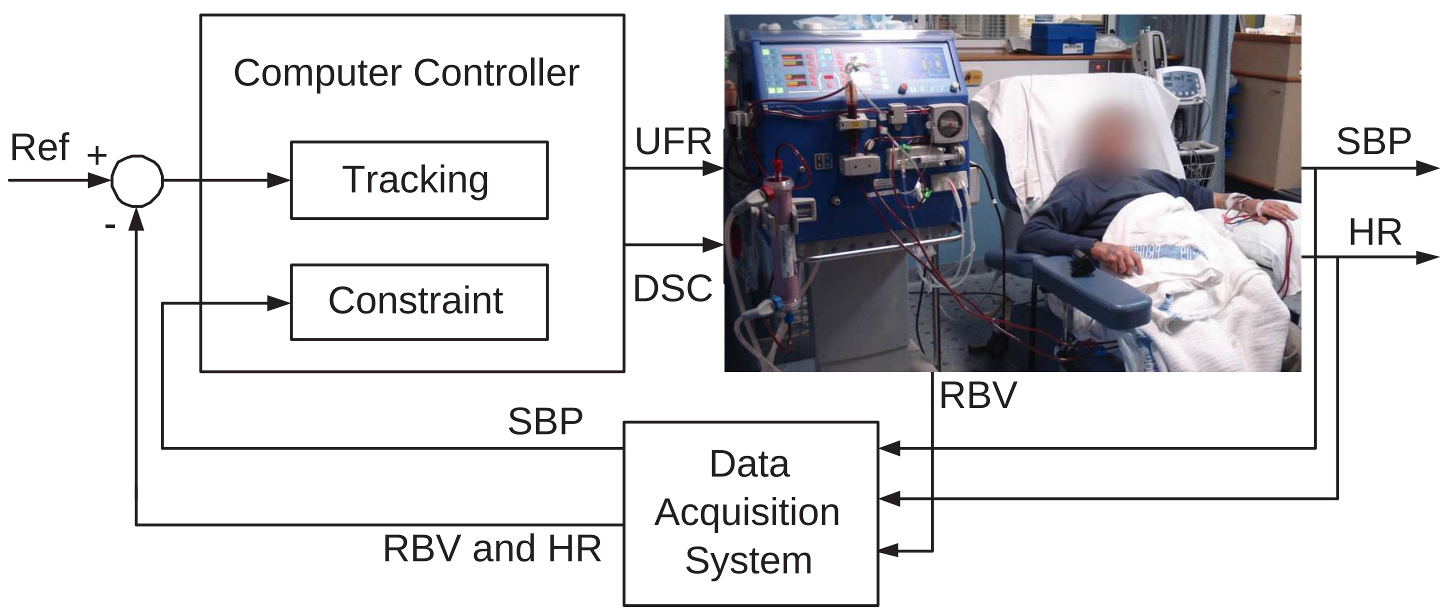


The controller

- Part 2: Model Predictive Controller Design

Computer-controlled hemodialysis systems

- The system regulates HR and RBV during hemodialysis as well as maintain SBP within stable range.



Controller design

- The designed computer-controller is based on MPC approach utilizing pre-defined constraints on the control inputs (UFR and DSC) as well as output (SBP).
- MPC predicts and optimize the future response of the system based on the dynamic model of the process.
- At each control interval, MPC solves an open loop control problem over some future intervals by taking into account current and future constraints.
- Minimizes an objective function and generates optimal control sequence
- The first control input is injected into the system and the same procedure is repeated at next control interval.

Controller design (Cont'd)

- First the system defined in (1) was re-arranged as:

$$\begin{aligned} \mathbf{x}_{k+1} &= \mathbf{A}_k \mathbf{x}_k + \mathbf{B}_k \mathbf{u}_k \\ \mathbf{y}_k &= \mathbf{C}_k \mathbf{x}_k \end{aligned} \tag{4}$$

$$A_k = \begin{bmatrix} (1-\xi) & 0 & 0 & a_2 \\ b_{1,k} & 1 & 0 & 0 \\ c_{1,k} & 0 & 1 & c_3 x_{3,0} - 0.5 c_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$B_k = \begin{bmatrix} \alpha a_{1,k} & -\beta a_{1,k} \\ 0 & 0 \\ 0 & c_2 \\ 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 & 1 & 0 \end{bmatrix},$$

Cost function

- The cost function is given by:

$$\mathbf{V_N^0} = \sum_{i=k+1}^{k+PH} \left[\left(\hat{x}_{1,k} - \eta_1 R_{1,k}\right)^2 + \left(\hat{x}_{2,k} - \eta_2 R_{2,k}\right)^2\right] + \sum_{i=k}^{k+M-1} \left(u_{1,k} - \eta_3 u_{2,k}\right)^2 \quad (5)$$

- It penalize the deviation of the states $x_{1,k}$ and $x_{2,k}$ from the reference trajectories $R_{1,k}$ and $R_{2,k}$ respectively.
- Another objective was to keep both the control inputs as close as possible to each other so that when a higher UFR is applied, the DSC is increased to ensure stability of patient.

Constraints

- The constraint on UFR is given by:

$$0 \leq u_{1,k} \leq UFR_{max} - (m/T_d)k;\ k = 0, 1, \ldots, M-1 \tag{6}$$

- The constraint on DSC is given by:

$$\begin{aligned}(0.5 - 0.5e^{-0.01k}) \leq u_{2,k} \quad \leq \quad & (0.5 + 0.5e^{-0.01k}); \\ & k = 0, 1, \ldots, M-1\end{aligned} \tag{7}$$

- Constraint was applied on SBP, introduced as the output constraint as:

$$SBP_{min} \leq y_k \leq SBP_{max};\ k = 0, 1, \ldots, P-1 \tag{8}$$

Constraints

- These constraints were expressed as linear constraints on U of the form:

$$LU \leq K \tag{9}$$

$$L = [I \quad -I \quad D \quad -D]^T \tag{10}$$

$$K = [\overline{c}_{1\ max} \quad \overline{c}_{2\ max} \quad \overline{c}_{1\ min} \quad \overline{c}_{2\ min} \quad \overline{c}_{3\ max} \quad \overline{c}_{3\ min}]^T \tag{11}$$

$$U = [u_{1,k} \quad \ldots \quad u_{1,k+M-1} \quad u_{2,k} \quad \ldots \quad u_{2,k+m-1} \quad]^T \tag{12}$$

where I is ($2M$x$2M$) identity matrix.

Constraints

- D is following ($P - 1$x$2M$) matrix:

$$D = \begin{bmatrix} CB & 0_{1x2} & \dots & 0_{1x2} \\ CAB & CB & \dots & 0_{1x2} \\ \vdots & \vdots & \ddots & \vdots \\ CA^M B & CA^{M-1}B & \dots & CAB \end{bmatrix}, \quad (13)$$

and

$$\begin{aligned} \overline{c}_{1\ max} &= [\overline{c}_{1,k\ max} \quad \dots \quad \overline{c}_{1,k+M-1\ max}]^T \\ \overline{c}_{2\ max} &= [\overline{c}_{2,k\ max} \quad \dots \quad \overline{c}_{2,k+M-1\ max}]^T \\ \overline{c}_{1\ min} &= [\overline{c}_{1,k\ min} \quad \dots \quad \overline{c}_{1,k+M-1\ min}]^T \\ \overline{c}_{2\ min} &= [\overline{c}_{2,k\ min} \quad \dots \quad \overline{c}_{2,k+M-1\ min}]^T \\ \overline{c}_{3\ max} &= [\overline{y}_{max} - CAx_0 \quad \dots \quad \overline{y}_{max} - CA^{P-1}x_0]^T \\ \overline{c}_{3\ min} &= [\overline{y}_{min} - CAx_0 \quad \dots \quad \overline{y}_{min} - CA^{P-1}x_0]^T \end{aligned} \quad (14)$$

Multiple Model Approach

- Considering the high variability between patients and the variability within the same patient on different days, it is inadequate to use a fixed model.
- A multiple model approach was utilized.
- At each sampling time k the model parameters were updated based on the parameterized model that best fits the system response over $k-1, k-2, \ldots, 1$. In mathematical terms, the following cost functions were solved:

$$\min_{\sigma(1,j)} \left(\sum_{i=1}^{k} (X_1(i) - \hat{X}_1(i,j))^2 + (X_2(i) - \hat{X}_2(i,j))^2 \right) \tag{15}$$

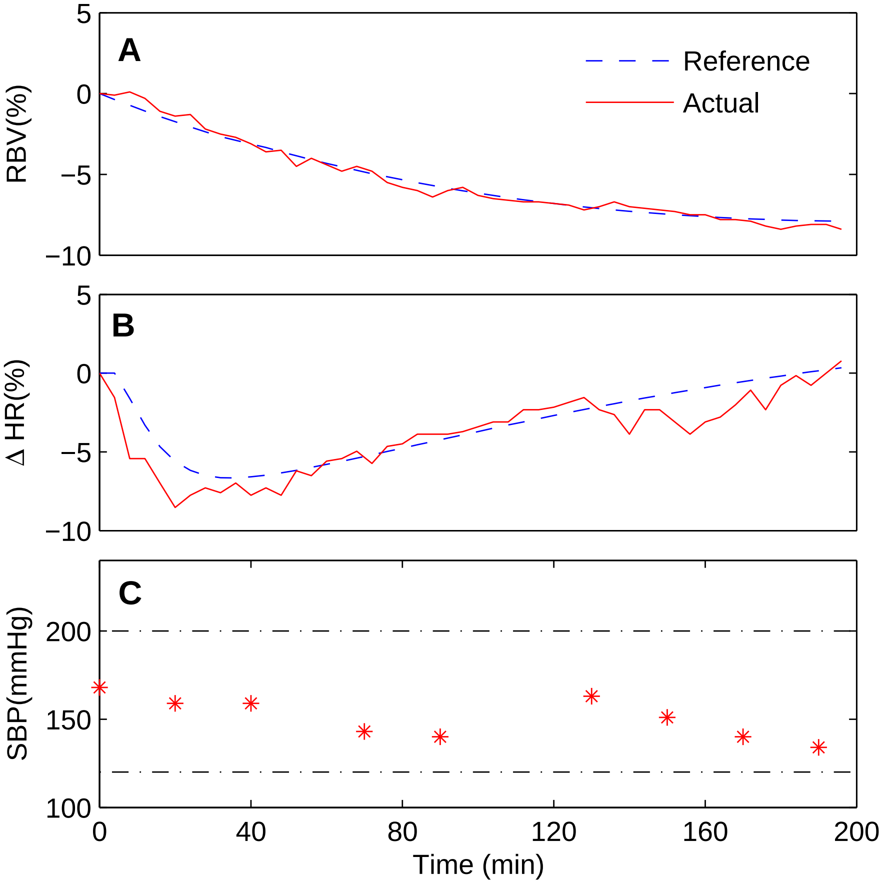
$$\min_{\sigma(2,j)} \left(\sum_{i=1}^{k} (X_3(i) - \hat{X}_3(i,j))^2 \right) \tag{16}$$

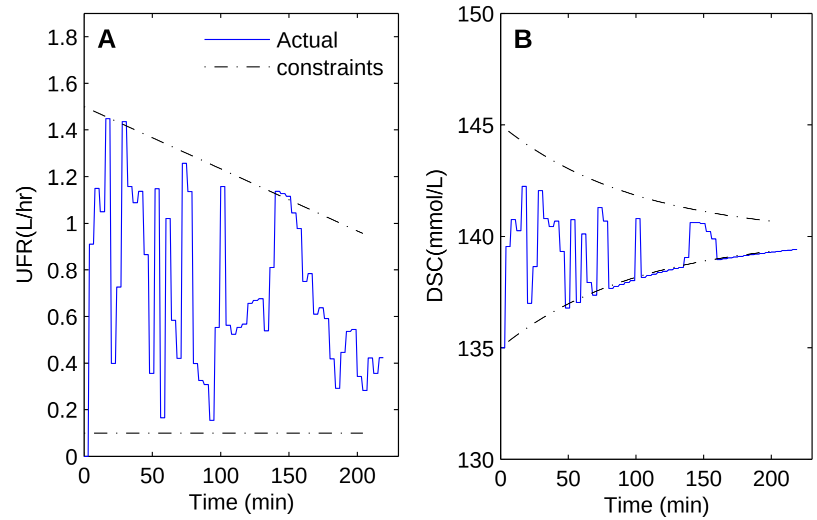
- The updated system parameters were then input to MPC in order to find the optimal control input that was used to find the future states.

Experimental results

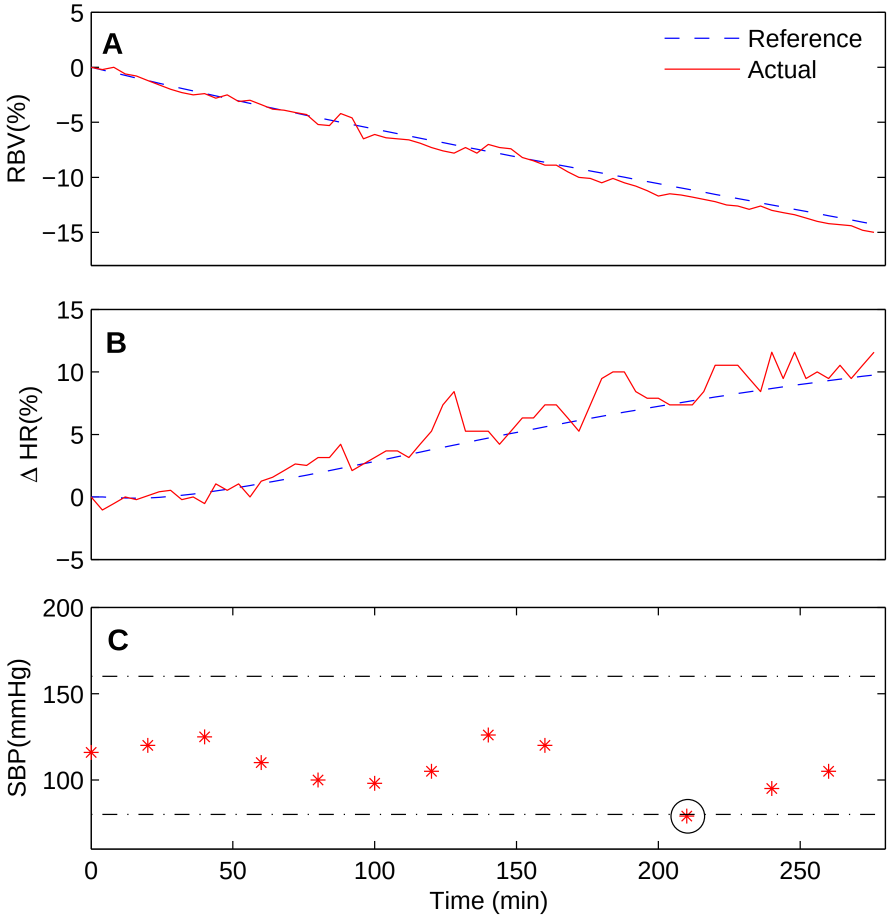
- To validate the developed system, it was tested on real patients from Prince of Wales Hospital, Sydney.
- Each patient initially underwent three consecutive dialysis sessions with constant UFR and DSC to determine the reference profiles for HR and RBV.
- These profiles were chosen based on these dialysis sessions as well as the suggestions of our clinical collaborators.
- The constraints on control inputs were based on recommendations from the nursing staff.

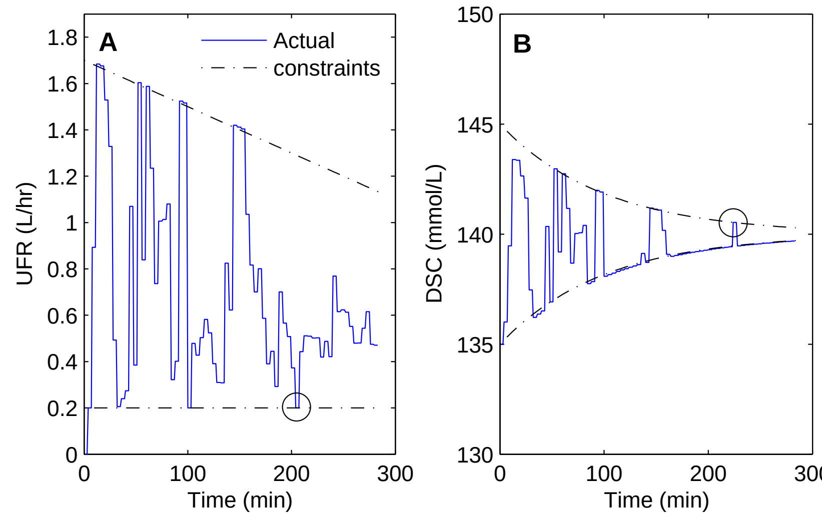
Results: Patient 1 (Hemodynamically stable)





Results: Patient 1 (Asymptomatic hypotensive)





Concluding remarks

- A linear time-varying state-space system describing the HR, SBP and RBV response to hemodialysis was proposed. The system uses the UFR and DSC as the control input.
- A Model predictive control based feedback system was designed and tested for the regulation of HR and RBV during hemodialysis while maintaining SBP within stable range.
- Model predictive control based approach was utilized that continuously adjust the UFR and DSC in order to track the changes in hemodynamic variables.
- Such system is a positive step towards developing new technologies capable of preventing dialysis induced complications.